

Lab Exercise

▪ Matrix-vector multiplication using multi-threading

- Product of **M*N matrix** and **N*1 vector** is **M*1 vector**.
- Row size (**M**) and column size (**N**) of matrix are given as command-line arguments.
- Elements of matrix and vector are randomly generated (data type is **uint64_t**)
 - » Reduce range to **0~9** with modular reduction (%) by using multiple threads.
 - » **thread_mod_reduc_data struct** is used to read the array pointer and size.
- Matrix-vector multiplication is done with multiple threads.
 - » **Each thread performs a dot product between a row and the vector.**
 - » **thread_mvm_data struct** is used to read the vector, matrix row, and column size (**N**) and store calculation result (**uint64_t**).
- Main function must wait for all threads to terminate before terminating.
- The multi-threaded operations are done in batches of **c** threads, where **c** is the number of cores in the system.

Lab Exercise Hint

- Matrix-Vector multiplication

$$\begin{array}{lcl} \text{Thread \#0} & \rightarrow & \\ \text{Thread \#1} & \rightarrow & \\ \text{Thread \#2} & \rightarrow & \end{array} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} ax + by + cz \\ dx + ey + fz \\ gx + hy + iz \end{bmatrix}$$

- Example

```
spl on  SPL in 2025s/w11/week11 via C v13.3.0-gcc
 ./w11_debug 10000 100
Writing matrix to file input_matrix.txt!
Writing matrix to file input_vector.txt!
Writing matrix to file result.txt!
Writing matrix to file answer.txt!
The result is correct!

spl on  SPL in 2025s/w11/week11 via C v13.3.0-gcc
 diff result.txt answer.txt
```

Lab Exercise Hint

- **Skeleton code and Makefile are provided.**
- **Provided Makefile compiles the program into two executables: w11 and w11_debug.**
- **w11_debug checks if the result matches with the result from the single-threaded function.**
- **`make test` checks if there are memory issues with the program.**
 - You may have to do ``sudo apt install valgrind``.

Exercise Submission

- **Submit your source code and Makefile in iCampus**
 - Bundle **source code** and **Makefile** with tar command
`$ tar caf [student_id]_w11.tar.zst week11`
 - We'll grade your submission with **make**
 - » If compilation fails, your points for this exercise will be zero